Practice workbook · Estimating an integral using a series expansion

Integrating the first few terms of a binomial expansion to estimate a definite integral.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-18

1. Question 5 · part d · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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1. Question 5 · part d · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d only.

Question 5 continued

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1. Question 5 · part d · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d only.

Question 5 continued

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1. Question 5 · part d · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part d only.

Question 5 continued

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(Total for Question 5 is 11 marks)

1. Question 5 · part d · 4 marks · Working and answer

Working and answer

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2. Question 9 · part e · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

9 (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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2. Question 9 · part e · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

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Question 9 continued

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2. Question 9 · part e · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

Question 9 continued

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2. Question 9 · part e · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

Question 9 continued

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(Total for Question 9 is 14 marks)

2. Question 9 · part e · 5 marks · Working and answer

Working and answer

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3. Question 10 · part d · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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3. Question 10 · part d · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 13 marks)

3. Question 10 · part d · 4 marks · Working and answer

Working and answer

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4. Question 7 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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4. Question 7 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 7 continued

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4. Question 7 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 7 continued

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4. Question 7 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 7 continued

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(Total for Question 7 is 11 marks)

4. Question 7 · part e · 3 marks · Working and answer

Working and answer

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5. Question 9 · part c · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

9 (a) Expand $\left(1-8x^2\right)^{\frac{1}{2}}$ in ascending powers of x, up to and including the term in x^6 giving each coefficient as an integer.

(3)

$$g(x)=\frac{a+bx}{\sqrt{1-8x^2}} \quad \text{where } a \text{ and } b \text{ are prime numbers}$$

Given that the fourth and fifth terms, in ascending powers of x, in the series expansion of $g(x)$ are $20x^3$ and $48x^4$ respectively,

(b) find the value of a and the value of b

(4)

Using the first five terms, in ascending powers of x, in the series expansion of $g(x)$

(c) obtain an estimate, to 4 significant figures, of $\int_0^{0.2} g(x)\,dx$

(4)

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5. Question 9 · part c · 4 marks

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Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 4 marks

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Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 4 marks

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Question 9 continued

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(Total for Question 9 is 11 marks)

5. Question 9 · part c · 4 marks · Working and answer

Working and answer

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